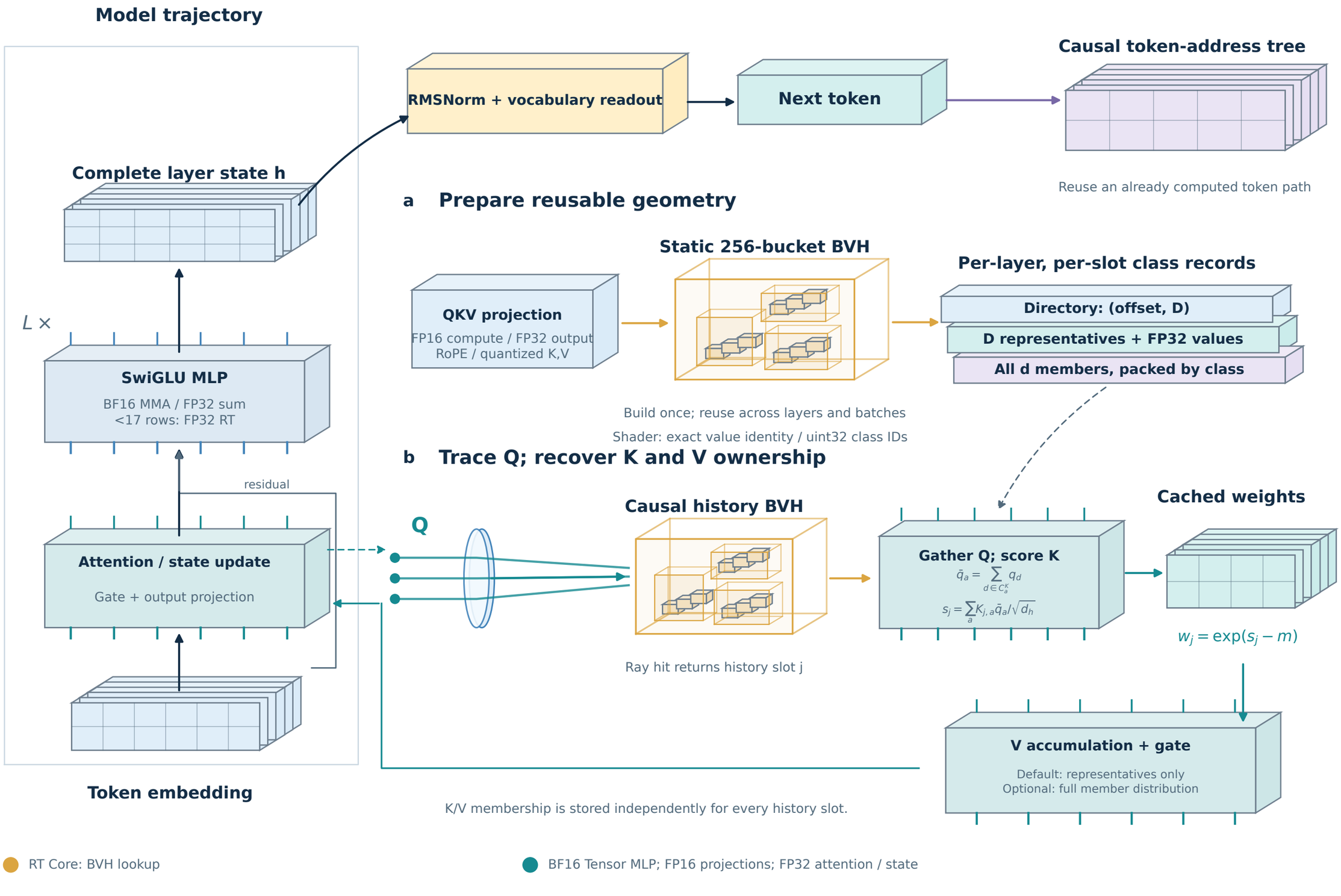
Ray Tracing Attention

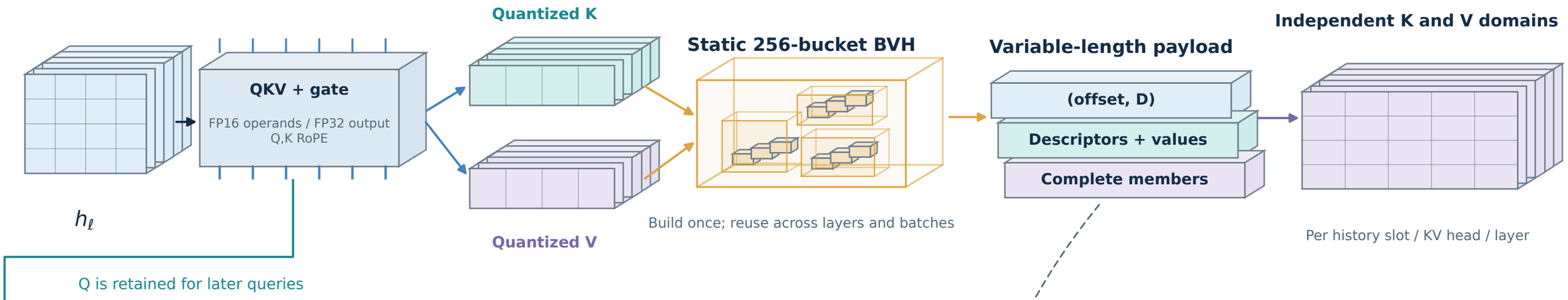
RT queries; BF16 MLP / FP32 sums; FP16 projections; FP32 attention and layer state



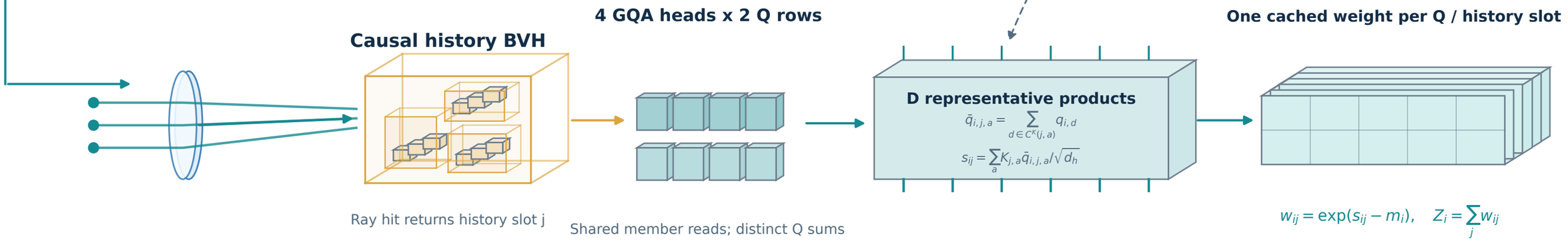
Attention: geometry, ownership and optical readout

FP16 projections produce FP32 signals; RT finds ownership; FP32 detectors combine full V contributions

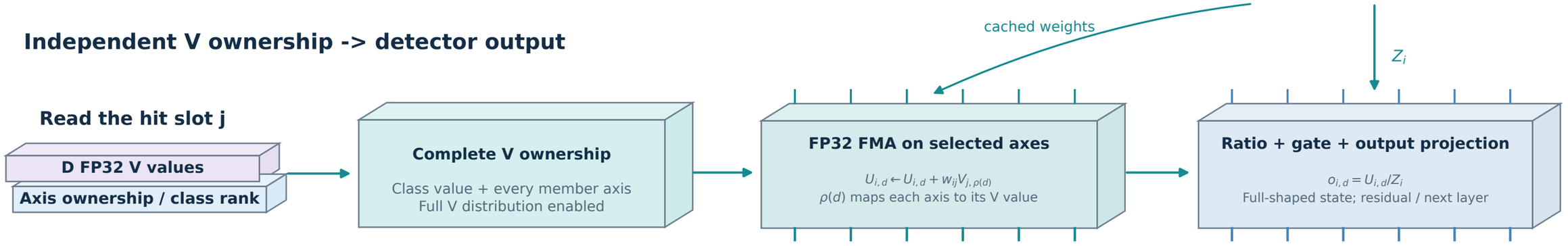
a Write K,V once; classify with a permanent scene



b RT history hit -> compact K gather -> cached weights



c Independent V ownership -> detector output



The measured configuration distributes each V-class value to every member axis.

RT Core: BVH lookup

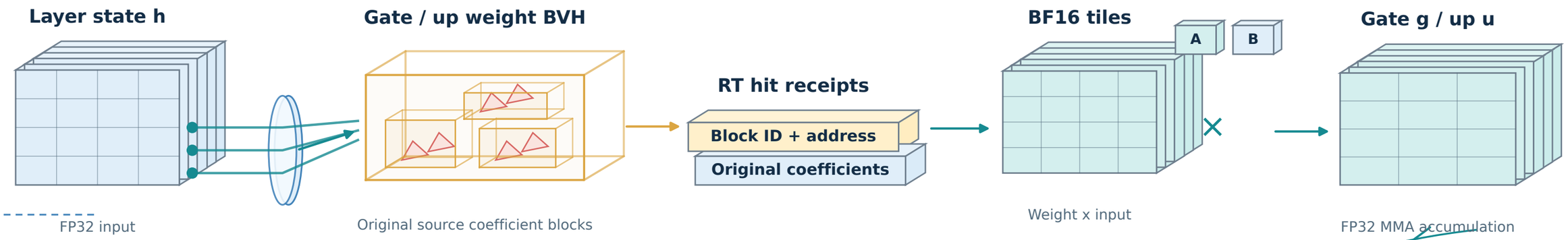
FP32 Shader: exact identity + score + weight + V accumulation

RT + BF16 Tensor MLP

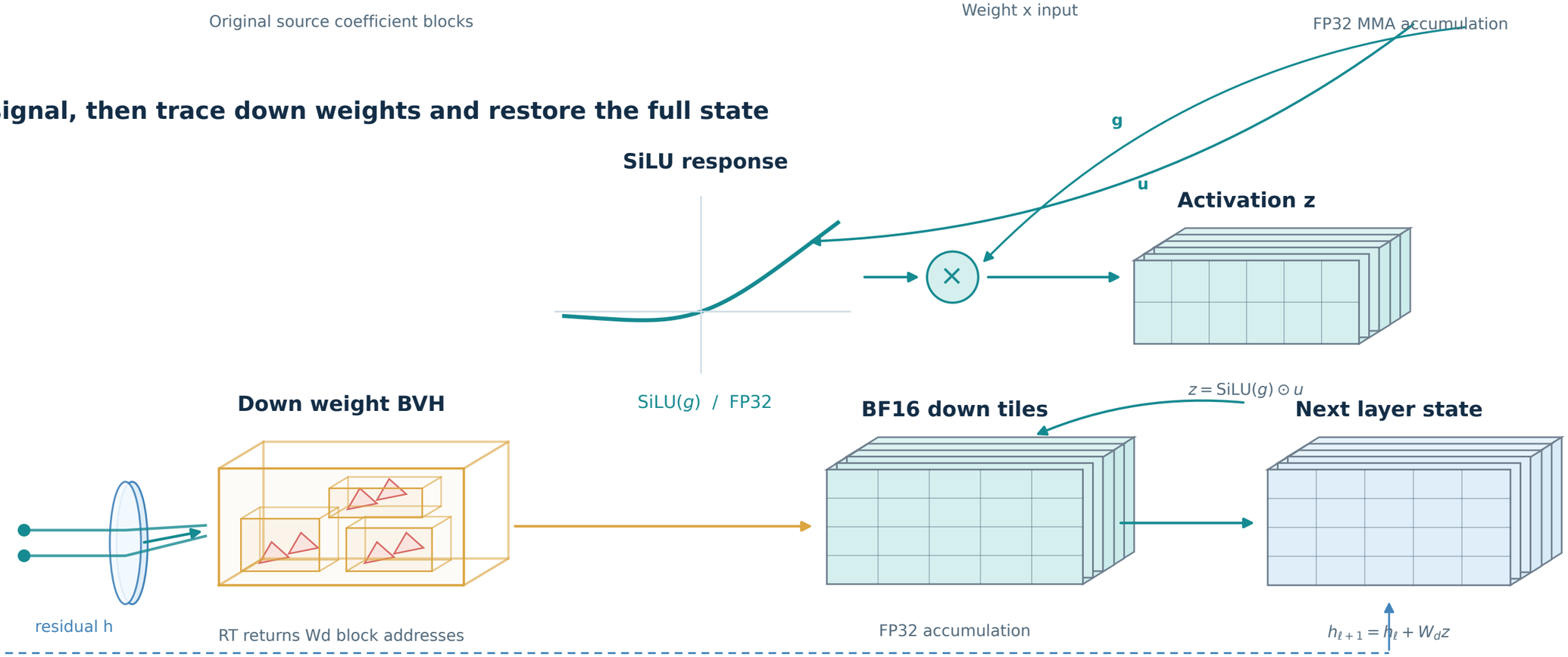
Immutable weight geometry supplies addresses; BF16 matrix products and FP32 nonlinear readout recover the next state

a Trace fixed weight blocks; multiply gate and up tiles

$$g = W_g h, \qquad u = W_u h$$



b Shape the signal, then trace down weights and restore the full state



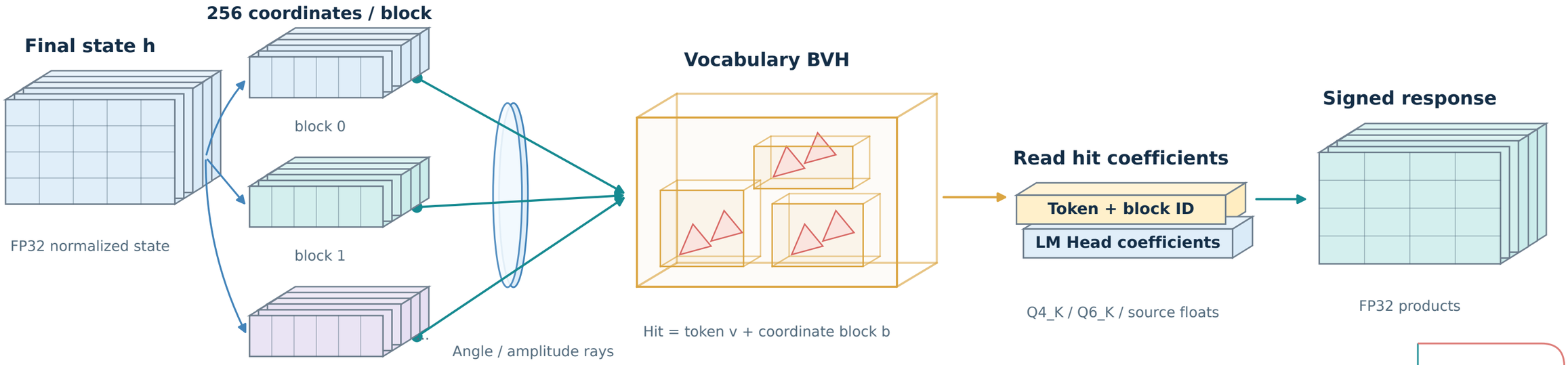
Batches >= 17 rows: BF16 Tensor products. Smaller batches and single-token decode: FP32 RT transmission.

High-dimensional state · RT: geometry / hit identity · Tensor: BF16 operands / FP32 sums

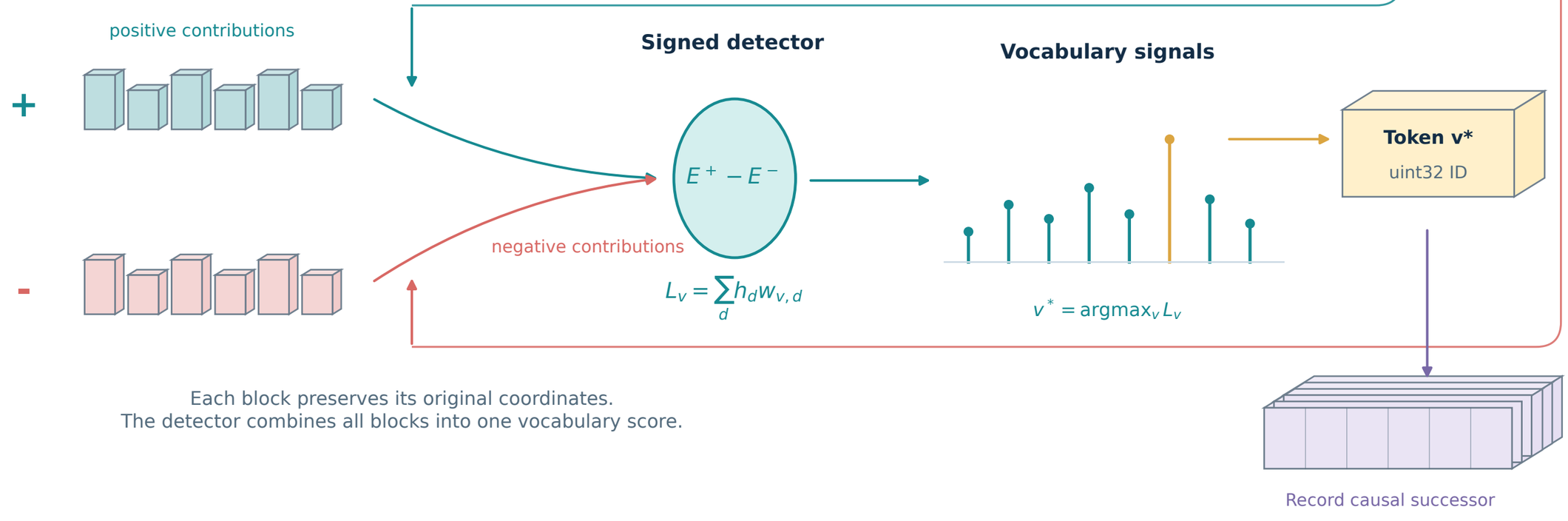
Vocabulary: geometry to the strongest optical signal

Full coordinate blocks retain their identities; original LM Head coefficients drive FP32 detector contributions

a Split the high-dimensional state; trace vocabulary blocks



b Separate signed channels; recombine and select the maximum



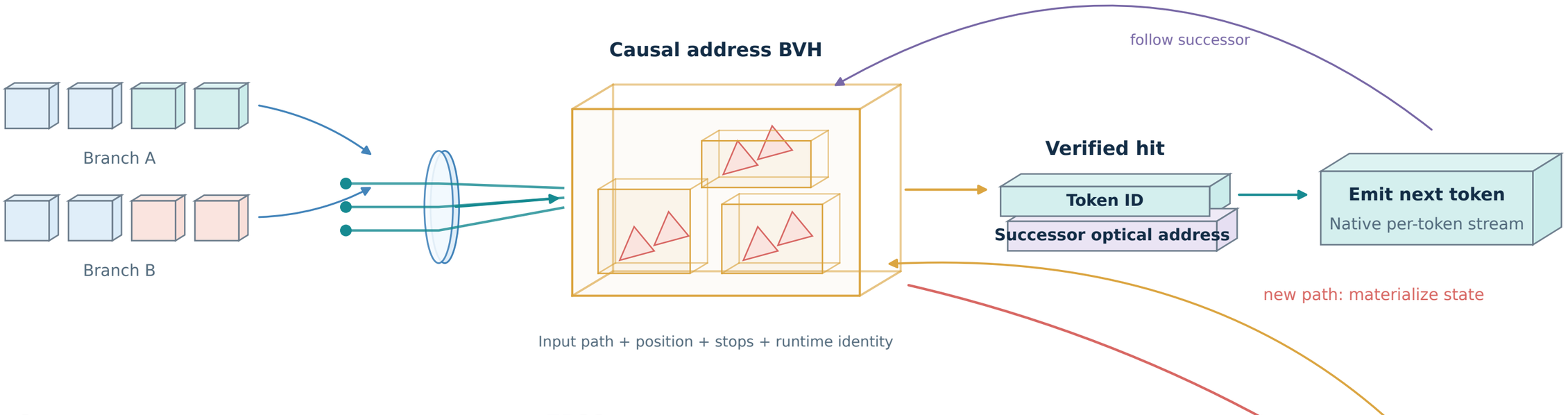
Each block preserves its original coordinates.
The detector combines all blocks into one vocabulary score.

High-dimensional state · RT: geometry / hit identity · Detector: FP32 contributions

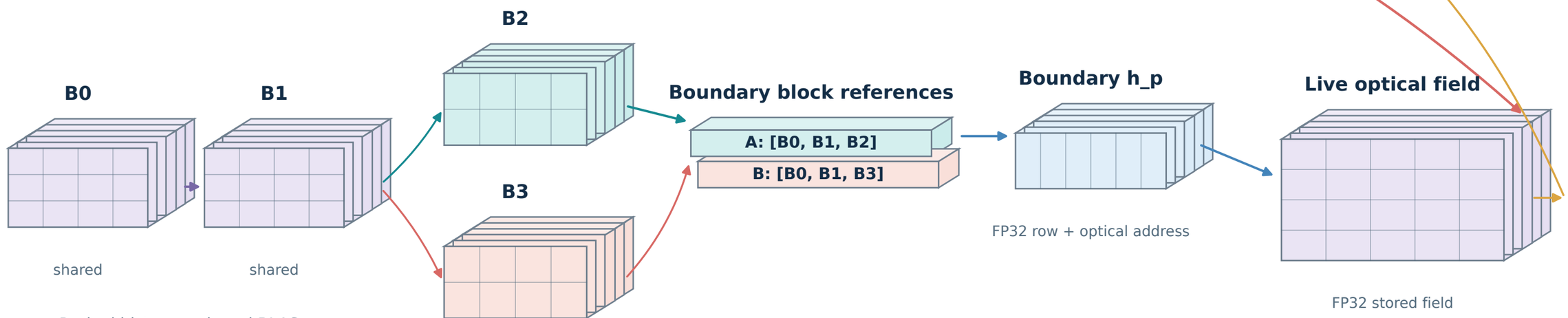
Causal reuse: shared geometry, compact boundaries

Matching token paths emit immediately; a new path materializes the optical field and records its successor

a Trace the causal address; follow token and successor hits



b Share immutable history blocks; retain an FP32 boundary row



Input and output endpoints share geometry; only the current recurrent field needs to be live.

High-dimensional state · RT: geometry / hit identity · Token + successor address